

PRANAV MATHAM

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EDUCATION

SJB Institute of Technology | BE, Computer Science and Engineering

GPA: 7.3 / 10 | Nov 2022 – Jun 2026

• Courses: Deep Learning, Artificial Intelligence, DBMS, Machine Learning, Analysis & Design of Algorithms, Object-Oriented Programming (Java), Cyber Security, Cloud Computing, Software Engineering & Project Management.

SKILLS

Programming Languages: Python, Java, C++, C, JavaScript, SQL

Frameworks & Tools: Django, Flask, FastAPI, React, Node.js, Flutter, Git/GitHub, VS Code, PyCharm

Technologies: Database Management Systems, REST APIs, SQL/NoSQL Databases (SQLite, Firebase), Data Analysis, Unit Testing, Debugging, WebSockets, SDLC, Agile/Scrum

Certifications: GitHub Copilot Fundamentals, Artificial Intelligence, Microsoft Excel & Power BI, Figma

Publication: “A Novel Framework for Automated Detection and Classification of Thyroid Nodules,” IEEE, 2026

WORK EXPERIENCE

Software Intern, Vectora Ventures Private Limited, Bengaluru, India

Aug 2026 – Present

- Assisting with software development tasks including coding, testing, debugging, documentation, and issue resolution in a hybrid work environment.
- **Key Skills:** Software Development, Debugging, Documentation, Agile/Scrum.

Python Full-Stack Developer Intern, Test Yantra Software Solutions, Bengaluru, India

Jan 2026 – Jun 2026

- Built the BusPass backend using Python, Django, and SQL, modeling student profiles, bus routes, and pass applications with relational database workflows for application and status tracking.
- Implemented REST/AJAX data flows and admin approval workflows; debugged frontend-backend integration issues and tested application behavior in an Agile/Scrum environment.
- **Key Skills:** Python, Django, SQL, DBMS, REST APIs, Unit Testing, Git/GitHub, Agile/Scrum.

Software Development Trainee, PYSPIDERS, Bengaluru, India

Dec 2025 – May 2026

- Built responsive web components using HTML, CSS, JavaScript, and Python, applying structured UI development and debugging across project builds.
- Collaborated in Agile workflows with a cross-functional team to resolve frontend-backend integration issues across multiple project builds.
- **Key Skills:** HTML, CSS, JavaScript, Python, UI Development, Agile/Scrum.

PROJECTS

“Thyroid Nodule Detection & Classification using Deep Learning” — [Thyroid Project](#)

- Built an end-to-end ultrasound image analysis pipeline using EfficientNet-B0, covering preprocessing, segmentation, feature extraction, classification, and tunable processing stages.
- Achieved ~95% model accuracy and added Grad-CAM heatmaps for explainable predictions; maintained relational scan history and diagnostic confidence data.

“Urban-Tech (NeuroSphere-X) – Smart City Traffic Platform” — [Urban-Tech Project](#)

- Built a real-time platform using FastAPI, Python, WebSockets, PyTorch, React, and Leaflet to process telemetry, traffic data, geospatial routes, and live dashboard events.
- Implemented data-driven threat and congestion analytics with dynamic heatmaps, emergency-vehicle telemetry, and real-time visual monitoring.

“Railway Ticket Reservation Management System” — [Railway Project](#)

- Built a Flutter/Dart reservation application using BLoC, REST/GraphQL APIs, Firebase, JSON parsing, Futures/Streams, and Hive/SQLite for ticket data handling.
- Designed database-backed booking and synchronization workflows with local caching and performance, load, and latency testing.

“AI-Powered Nutrition Tracking Website – FoodieAI” — [Foodie Project](#)

- Built a full-stack nutrition platform using React, Node.js, Flask, and REST APIs with OpenCV, PIL, and NumPy to process food images and integrate USDA nutrition data.